Practical Session

Practical 1: Identify Your Network Information

Command:

```
ipconfig
```

Task:

- Find IPv4 Address
- Find Subnet Mask
- Find Default Gateway

Learning:

Students will understand how a computer gets a network address from a router.

Practical 2: Detailed Network Information

Command:

```
ipconfig /all
```

Task:

Identify:

- Computer Name
- MAC Address (Physical Address)
- DHCP Enabled Status
- DNS Server

Learning:

Students learn how devices are uniquely identified on a network.

Practical 3: Test Internet Connectivity

Command:

```
ping google.com
```

Task:

Observe:

- Reply from server
- Time (ms)
- Packets Sent/Received

Learning:

Understand how to test if the internet connection is working.

Practical 4: Continuous Ping Monitoring

Command:

```
ping -t google.com
```

Stop:

```
Ctrl + C
```

Learning:

Monitor network stability in real time.

Practical 5: View Saved Wi-Fi Networks

Command:

```
netsh wlan show profiles
```

Task:

List all Wi-Fi networks previously connected to the system.

Learning:

Understand how Windows stores wireless network profiles.

Practical 6: Display Connected Network Information

Command:

```
netsh wlan show interfaces
```

Task:

Find:

- Wi-Fi Name (SSID)
- Signal Strength
- Radio Type
- Connection State

Learning:

Understand wireless network connection details.

Practical 7: Check Active Network Connections

Command:

```
netstat
```

Learning:

See active network sessions and communication.

Practical 8: View Network Adapter Details

Command:

```
getmac
```

Task:

Find MAC Address of Ethernet and Wi-Fi adapters.

Learning:

Understand hardware identification in networking.

Hands-On Activity 1: Identify Networking Hardware

Provide actual devices and ask students to identify:

- Router
- Switch
- Access Point
- Ethernet Cable
- Fiber Cable (if available)
- RJ45 Connector

Task:

Write the name and purpose of each device.

Hands-On Activity 2: Cable Identification Challenge

Show different cables:

- HDMI
- USB
- Ethernet
- VGA

Task:

Students identify:

- Cable Name
- Purpose
- Network Cable or Not

Hands-On Activity 3: Build a Basic Network Diagram

Students draw:

```
Internet
   ↓
 Modem
   ↓
 Router
   ↓
 Switch
   ↓
 PC1   PC2   Printer
```

Learning:

Understand network topology.

Networking Tips & Tricks

Tip 1: Find Your Computer Name

```
hostname
```

Tip 2: Find Default Gateway Quickly

```
ipconfig | find "Default Gateway"
```

Tip 3: Check DNS Servers

```
nslookup google.com
```

Tip 4: Display Routing Table

```
route print
```

Shows how data travels through the network.

Tip 5: Flush DNS Cache

```
ipconfig /flushdns
```

Useful when websites are not opening correctly.

Tip 6: Renew IP Address

```
ipconfig /release
ipconfig /renew
```

Useful for troubleshooting network connectivity issues.

Mini Troubleshooting Task

Scenario:

Internet is not working on a PC.

Student Checklist:

1. Check Ethernet cable connection.
2. Check router power.
3. Run:
4. `ipconfig`
5. Ping:

6. `ping 127.0.0.1`
7. Ping:
8. `ping google.com`
9. Identify whether the problem is:
    - Cable issue
    - Router issue
    - ISP issue
    - DNS issue